42. During transcription holoenzyme RNA polymerase binds to a DNA sequence and the DNA assumes a saddle like structure at that point. What is that sequence called (JIPMER 2007)

a) CAAT box b) GGTT box
c) AAAT box d) **TATA box**

43. The successive nucleotides of RNA are covalently linked through (JIPMER 2001)

a) Hydrogen bonds
b) **Phosphodiester bonds**
c) Glycosidic bonds d) None of these

44. The Okazaki fragments in DNA chain growth (AIPMT 2007, JIPMER 2004)

a) Polymerize in the 3' - to 5' direction and forms replication fork
b) Prove semi conservative nature of DNA replication
c) **Polymerize in the 5' to 3' direction and explains 3' – to – 5' DNA replication**
d) Result in transcription

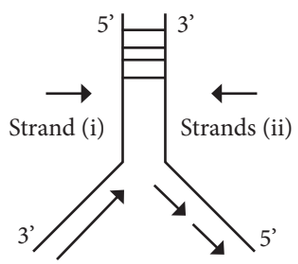
45. Taylor conducted the experiment to prove semiconservative mode of chromosome replication on (NEET (Phase II) 2016)

a) Drosophila melanogaster b) e-coli
c) Vinca rosea d) **vicia faba**

46. The new strand synthesized in small pieces and then joined together during DNA replication is called (AIIMS 1994)

a) Dead strand b) **Lagging strand**
c) Leading strand d) All of these

47. What is incorrect about the following figure representing DNA replication (AIIMS 2009)



a) The direction of DNA replication in strand (i)
b) The direction of DNA replication in strand (ii)
c) **Discontinuous replication of strand (i)**
d) Discontinuous replication of strand (ii)

48. DNA multiplication is called (JIPMER 2009)

a) Translation b) **Replication**
c) Transduction d) Transcription

49. The complete set of chromosome inherited as a single unit from one parent is known as (AIIMS 1994)

a) **Genome** b) Linkage
c) Gene pool d) Genotype

50. The mobile genetic element is (JIPMER 2014)

a) **Transposon** b) Mutation
c) Endonuclease d) Variation

UNIT VIII – Biotechnology

1. What is the criterion for DNA fragments movement on agarose gel during gel electrophoresis? (NEET 2017)

a) **The smaller the fragment size, the farther it moves.**
b) Positively charged fragments move to farther end.
c) Negatively charged fragments do not move.
d) The larger the fragment size, the farther it moves.

2. Stirred-tank bioreactors have been designed for (NEET – II 2016)

a) Purification of product.
b) Addition of preservatives to the product
c) **Availability of oxygen throughout the process**
d) Ensuring anaerobic conditions in the culture vessel.

3. Which of the following is not a component of downstream processing? (NEET-II 2016)

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